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Measurement of the laminar burning velocity and kinetics study of the importance of the hydrogen recovery mechanism of ammonia/hydrogen/air premixed flames



Gabriel J. Gotama^{a,b,c,*}, Akihiro Hayakawa^a, Ekenechukwu C. Okafor^d, Ryuhei Kanoshima^{a,e}, Masao Hayashi^{a,e}, Taku Kudo^a, Hideaki Kobayashi^a

^a Institute of Fluid Science, Tohoku University, 2-1-1 Katahira, Aoba-ku, Sendai, Miyagi 980-8577, Japan

^b School of Mechanical and Aerospace Engineering, Nanyang Technological University, 50 Nanyang Ave., 639798 Singapore

^c Department of Aerospace and Geodesy, Technical University of Munich, Willy-Messerschmitt-Str. 1, Taufkirchen/Ottobrunn 82024, Munich, Germany

^d Fukushima Renewable Energy Institute, National Institute of Advanced Industrial Science and Technology (AIST), 2-2-9 Machiikeda, Koriyama, Fukushima 963-0298, Japan

^e Department of Aerospace Engineering, Tohoku University, 6-6-1 Aoba, Aramaki, Aoba-ku, Sendai, Miyagi 980-5879, Japan

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ABSTRACT

The application of ammonia (NH₃) blended with hydrogen (H₂) as a fuel in combustion systems is a practical approach to decarbonise the energy sector, and the combustion of the fuel at rich conditions is relevant in emissions control through rich-lean combustion. However, the chemistry of rich NH₃/H₂ flames at high pressure, and the interaction between NH₃ and H₂ still need to be clarified. Therefore, the present study focuses on the chemical kinetics of NH₃/H₂/air flames at rich conditions and elevated pressures. To validate chemical kinetics in the literature, the laminar burning velocity of NH₃/H₂/Air premixed flames were measured at 0.1 and 0.5 MPa and equivalence ratios up to 1.8. The results show that the seven kinetics mechanisms studied could not satisfactorily predict the measurements at fuel-rich conditions and elevated pressure. The kinetics mechanism by Han et al. was optimized, leading to a new detailed kinetics, which can be reduced to 26 species and 119 reactions and satisfactorily predicts the present measurements and those in the literature. Analysis of the chemistry of NH₃/H₂ flames using the new mechanism shows NH₃ and H₂ kinetics are strongly coupled through a H₂ decomposition/recovery mechanism, here named H₂ recovery mechanism, which is important in modelling the burning velocity of the flame at fuel-rich conditions. The burned gas Markstein length was also extracted from the measured flame speed and its behaviour was studied using theoretical correlations.

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1. Introduction

Ammonia has recently gained interest as a carbon-free fuel for direct decarbonisation of the energy sector. Ammonia has various advantages as a fuel, such as its easy storage and transport, favourable energy density, and sustainable production [1–4]. Nevertheless, there are shortcomings in the fuel properties of ammonia which include the slow laminar burning velocity, low calorific value, narrow flammability limit, and large NO_x emissions that tends to hinder its practical use [5–7]. Several studies have suggested methods to circumvent these drawbacks in ammonia

combustion characteristics through unique combustor designs for pure ammonia flames or by blending ammonia with other fuels such as hydrocarbons and hydrogen which have higher flame speeds [8–10].

The blending of hydrogen with ammonia is considered one of the most promising ways to burn ammonia mixtures in internal combustion engines. The resulting carbon-free binary fuel has been reported to have higher laminar burning velocity [11], increased reactivity [12], and a larger flammability limit [9] than pure ammonia flames. However, the binary fuel may emit higher levels of NO_x than pure ammonia flames owing to the high flame temperature and the abundance of O, H, and OH radicals that promote fuel NO formation [6,13–15]. Since it is important to maintain acceptable level of NO_x, several works have been dedicated to understand and control NO formation in ammonia flames [14,16,17]. One proposed solution is to implement the two-stage rich-lean combustion

* Corresponding author at: Department of Mechanical Engineering, The University of Melbourne, Parkville, Victoria 3010, Australia
E-mail address: ggotama@student.unimelb.edu.au (G.J. Gotama).

which has been demonstrated in micro gas turbines to control the emissions of NO_x from ammonia containing mixtures [18–20]. This NO_x control strategy leverages the low NO production in fuel-rich ammonia flames. It is therefore important to clarify the chemical kinetics of rich NH_3/H_2 flames especially at elevated pressure conditions relevant to applications for the purpose of advancing our understanding of emission control mechanisms and the design of low emission combustors for the fuel.

A commendable amount of research effort has recently been spent on the development and optimization of chemical kinetics of ammonia blended fuels [21,22]. The optimization of the chemistry of these binary fuels requires an understanding of the chemistry of each component fuel and any relevant interactions between them. Okafor et al. [23] investigated the laminar burning velocity of $\text{NH}_3/\text{CH}_4/\text{air}$ flames and reported that the ammonia oxidation pathway does not have a significant direct interaction with methane oxidation pathway which may influence the burning speed. An indirect coupling of the two kinetics was found to be through their dependence on the same pool of O, H and OH radicals. This suggests that a combination of an optimized methane oxidation kinetics with that of ammonia may produce an oxidation kinetics for NH_3/CH_4 binary fuel, which may not require any further optimization. Arunthanayothin et al. [24] further investigated the effects of ammonia addition on the oxidation of methane in a jet-stirred reactor and found that ammonia has a considerable effect on methane oxidation at low to intermediate flame temperature. At higher temperature, there are fewer direct interactions between the oxidation steps of these fuels.

On the other hand, some studies have found significant interactions between the NH_3 and H_2 oxidation kinetics in $\text{NH}_3/\text{H}_2/\text{air}$ flames. Otomo et al. [15] investigated H_2 formation in $\text{NH}_3/\text{H}_2/\text{air}$ flames and found a mechanism of H_2 consumption and regeneration. They suggest that $\text{O} + \text{H}_2 = \text{H} + \text{OH}$ and $\text{OH} + \text{H}_2 = \text{H} + \text{H}_2\text{O}$ contribute to H_2 consumption, while certain reactions between H radicals and NH_x result in H_2 regeneration and encourage NH_3 decomposition. They also found that reactions between H radical and HNO or N_2H_2 contribute to H_2 formation. The direct coupling between the H_2 oxidation pathway and the NH_3 oxidation pathway may become more significant in flame with sufficiently high concentrations of both fuels and may be strengthened in rich ammonia flames because of the high H_2 content and the importance of the N_2H_2 chemistry in rich ammonia flames. However, the kinetics of the interaction between H_2 and NH_3 chemistry has not been given much attention especially at fuel rich conditions and elevated pressures.

The deficiency in modelling the kinetics of the direct coupling between the NH_3 and H_2 oxidation pathway may contribute to the inability of existing reaction models to satisfactorily predict the burning velocity of NH_3/H_2 flames at some conditions. Da Rocha et al. [25] assessed the performance of ten mechanisms in predicting the laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ flames and noted that the results from the models were scattered. Following that study, Han et al. [26] developed a reaction mechanism that can satisfactorily predicts the laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ flames over a relatively wide range of conditions. However, a subsequent study by Wang et al. [27] showed that the mechanism by Han et al. [26] underpredicts the laminar burning velocity for lean condition at 0.1 MPa and overpredicts the laminar burning velocity for rich condition at 0.3 and 0.5 MPa.

Therefore, this study aims to clarify and optimize the chemistry of the interaction between NH_3 and H_2 oxidation pathways at fuel-rich conditions and elevated pressures. In order to validate existing chemical kinetics models at the conditions of interest in this study, the unstretched laminar burning velocity of mixtures of $\text{NH}_3/\text{H}_2/\text{air}$ flames were measured. Although various studies have

reported the laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{air}$ flames at various conditions as shown in Table 1, there is scarcity of data at rich equivalence ratios up to 1.8 at 0.5 MPa.

While the laminar burning velocity has been the focus in the combustion characteristics studies of $\text{NH}_3/\text{H}_2/\text{Air}$ flames, the burned gas Markstein length has not received similar attention. Markstein length provides a quantitative evaluation of the sensitivity of the flame speed to stretch [33], and has been associated with the turbulent flame characteristics [34,35]. In addition, the Markstein length may correlate with the onset of flame instability due to hydrodynamic and thermo-diffusive effects. However, there are only few studies on the Markstein length and Markstein number (normalised Markstein length) of $\text{NH}_3/\text{H}_2/\text{Air}$ flames such as the studies by Ichikawa et al. [11], Lee et al. [28,29], and Li et al. [13], as shown in Table 1. Ichikawa et al. [11] measured the Markstein length of stoichiometric flames at various pressures, while Lee et al. [28,29] and Li et al. [13] conducted their investigation at 0.1 MPa. The variation of the Markstein length with equivalence ratio at high pressure has not yet been reported and should be investigated.

Therefore, with the motivation to improve the modelling and the understanding of rich ammonia/hydrogen flames at high pressure, the present study investigated the laminar burning velocity and Markstein length of $\text{NH}_3/\text{H}_2/\text{Air}$ flames for various conditions using the spherical flame method, numerical modelling, and theoretical evaluation. The measured laminar burning velocity was used to develop a reaction mechanism that can satisfactorily model the present measurements. The developed reaction mechanism was reduced to lower the computational cost. Several analyses were conducted using the new mechanism to investigate the flame chemistry with interests in the interaction between NH_3 and H_2 .

2. Experimental method

In this study, a constant volume combustion chamber with an inner diameter of 0.27 m, length of 0.41 m, and volume of 0.023 m^3 was used. A similar setup has been previously used by Ichikawa et al. [11]. The mixture of dry air, NH_3 (purity > 99.9%), and H_2 (purity > 99.9%) were supplied to the chamber. The mixture was ignited using a pair of spark electrodes with a diameter of 1.6 mm. The electrodes were positioned at the centre of the chamber with a gap of 2 mm between them. The spherically propagating flames were visualized using the Schlieren technique and recorded using a high-speed camera (Photron, FASTCAM Mini AX100) with macro-lens (Nikon, Ai AF Micro-Nikkor 200 mm f/4 IF-ED) at a resolution of 768×768 pixels.

Mixtures with equivalence ratios, ϕ , of 0.8 to 1.8 were investigated under initial pressures, P_i , of 0.1 and 0.5 MPa. The initial temperature, T_i , of the mixture was set to 298 ± 5 K. The H_2 ratio in the binary fuel, X_{H_2} , is defined in Eq. (1) with the symbol [i] representing the mole fraction of the 'i' species. X_{H_2} was set to 0.4 in the present study as mixtures of $\text{NH}_3/\text{H}_2/\text{Air}$ flames with $X_{\text{H}_2} = 0.4$ has a similar laminar burning velocity as methane/air premixed flames at stoichiometry [11].

$$X_{\text{H}_2} = \frac{[\text{H}_2]}{[\text{NH}_3] + [\text{H}_2]} \quad (1)$$

To enable ignition of the mixture with $\phi = 1.8$ at 0.1 MPa, electrostatic energy of 9.4 J was charged in the CDI capacitor (300 μF , 250 volts) and used at all conditions for consistency. Mixtures with $\phi = 2.0$ were also ignited however large buoyancy effects (eccentric flame propagation) could not allow accurate measurement of the laminar burning velocity.

In this study, the initial pressure was measured using a GE UNIK 500 silicon pressure sensor with improved accuracy of 0.02%

Table 1Experimental studies for the laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames.

Year	Author(s)	Method*	H_2 ratio in the binary fuel	Equivalence ratio	Initial temperature (K)	Initial pressure (MPa)
2010	Lee and Kim et al. ^a [28]	CVCC	0–0.5	0.6–1.67	298	0.1
2010	Lee, Lee and Kwon ^a [29]	CVCC	0.7–1.0	0.6–1.67	298	0.1
2013	Kumar and Meyer [30]	CF	0.247–1.0	0.5–1.1	298	0.1
2014	Li et al. ^a [13]	CF	0.34–0.6	0.6–1.4	298	0.1
2015	Ichikawa et al. ^a [11]	CVCC	0–1.0	1.0	298	0.1–0.5
2019	Han et al. [31]	HF	0–0.45	0.7–1.6	298	0.1
2020	Lhuillier et al. [32]	CVCC	0–0.6	0.8–1.4	298–473	0.1
2020	Wang et al. [27]	HF	0.4 and 0.6	0.6–1.6	298	0.1–0.5
2021	Shrestha et al. [22]	CVCC	0–0.3	0.8–1.4	473	0.3–0.7
2021	Present study ^a	CVCC	0.4	0.8–1.8	298	0.1, 0.5

*CVCC: Constant volume combustion chamber; CF: Conical flame; HF: Heat flux.

^a Reported the burned gas Markstein length / Markstein number.

FS (full scale). Previous studies [5,23,36] using a similar setup have shown that the pressure in the chamber remained close to the initial pressure during the recorded period. The pressure does not significantly change since the diameter of the optical window (60 mm) is sufficiently small compared to the chamber (270 mm). Furthermore, because of the small ratio between the diameters of the optical window and the chamber, confinement had negligible influence on the measurements. According to Burke et al. [37], the confinement effect can be neglected when the ratio between the flame radii and the chamber is less than 0.3. In this study, the largest ratio between the flame radii and the chamber is 0.22.

The flame speed, S_n , was calculated using Eq. (2) by evaluating the change of radius, r_{sch} , with time, t , using the recorded flame propagation.

$$S_n = \frac{dr_{sch}}{dt} \quad (2)$$

Since the flames experienced stretching due to curvature, the total flame stretch rate, ε , was evaluated using Eq. (3) as the normalised change in the flame front area, A_f , with time.

$$\varepsilon = \frac{1}{A_f} \frac{dA_f}{dt} = \frac{2}{r_{sch}} \frac{dr_{sch}}{dt} \quad (3)$$

Linear [38] and non-linear [39] extrapolation methods, as given in Eqs. (4) and (5), respectively, were employed to obtain the unstretched flame speed, S_s , and the burned gas Markstein length, L_b . In this study, it was found that the non-linear extrapolation is the appropriate method to measure the laminar burning velocity because the Lewis number, Le , for some conditions reached far from unity [40]. This is discussed in more details in Section 4.4.

$$S_s - S_n = L_b \varepsilon \quad (4)$$

$$\left(\frac{S_n}{S_s}\right)^2 \ln\left(\frac{S_n}{S_s}\right)^2 = -2\left(\frac{L_b \varepsilon}{S_s}\right) \quad (5)$$

The unstretched laminar burning velocity, S_L , was calculated using Eq. (6) that relates the unstretched flame speed with the ratio of burned gas density, ρ_b , and unburned gas density, ρ_u . The densities were obtained through equilibrium calculation in ANSYS CHEMKIN-PRO 2020 R2 [41] using a mechanism by Han et al. [26].

$$S_L = S_s \frac{\rho_b}{\rho_u} \quad (6)$$

The Schlieren images of $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames at 0.1 and 0.5 MPa are shown in Figs. 1 and 2, respectively. Note that the brightness and contrast of the images have been adjusted for clarity. At 0.1 MPa (Fig. 1), a slight buoyancy effect was observed at $\phi = 1.8$. However, it did not alter the flame shape and considered not to have affected the measurements [23,36,42]. On the

other hand, at 0.5 MPa (Fig. 2), cellular instability was observed at lean conditions. Cellular instability results in an increase in the stretched flame speed due to an increase flame area and leads to large inaccuracy in measured unstretched laminar burning velocity [33]. On the other hand, at the same pressure condition, flames with $\phi = 1.6$ and 1.8 were convected upwards from the middle of the chamber due to buoyancy effects. However, they maintained an approximately spherical shape. The buoyancy effect originated from the slow laminar burning velocity of the flames (See Section 4.1). Nevertheless, the flame speed is still sufficient to maintain the spherical shape of the flame and therefore the buoyancy effect on the measured flame speed was considered negligible [23,36,42].

In ensuring the accuracy of the measurements, several procedures were employed as follows:

- After every run, water vapours were produced in the chamber as a by-product of combusting ammonia/hydrogen binary fuel. The presence of water in the chamber can lead to ammonia's adsorption and alter the mole fraction of the mixture for the subsequent experiment [36]. To avoid ammonia's adsorption, the chamber was dried after every run. At 0.1 MPa, the chamber was dried by supplying air into the chamber. At 0.5 MPa, larger amount of water vapours were produced after each run, owing to the increased amount of fuel supplied compared to that at 0.1 MPa. Supplying air into the chamber was not sufficient to remove the water during the high pressure experiments and the chamber was opened after every run to be thoroughly dried.
- Five replications were conducted for each condition to ensure the significance of the results. The final measurements were based on the average of the various replications. Two standard deviations of the results were used as the error bar that corresponds to a 95% confidence interval. The uncertainty takes into account instability effects that may influence the variance of the measurements.
- The obtained flame radii have a broad data scatter and noise. In this study, the noise was attenuated using moving average filtering and adjusting the recording frame speed. The rate of filtering was regulated to avoid affecting the results.
- As shown in Fig. 3, only data points within the quasi-steady region were extrapolated to obtain the laminar burning velocity. The quasi-steady region corresponds to the flame speed that is minimally influenced by various effects such as ignition and cellular instability effects. The ignition effect was excluded using data points with $r_{sch} \geq 10$ mm [23,28,29] (as seen in Fig. 3) since the flame speed becomes independent from the ignition energy above a certain radius [43]. The cellular instability effect, indicated by a significant gradient change in the flame speed – stretch rate curve [33] (as seen in Fig. 3), was also excluded.

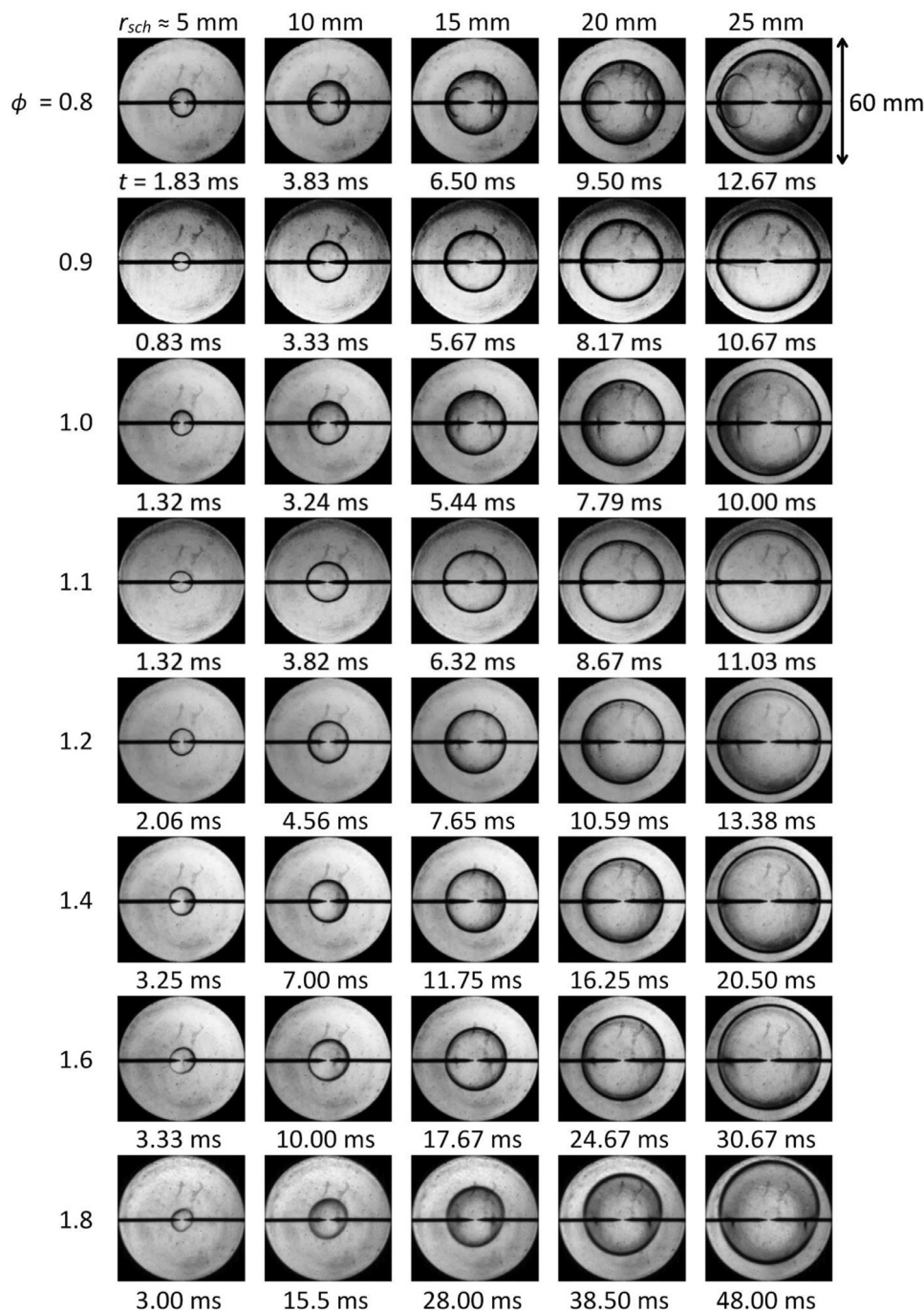


Fig. 1. Schlieren images of $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames for $X_{\text{H}_2} = 0.4$ at 0.1 MPa.

3. Numerical method and development of the reaction mechanism

A steady one dimensional, freely propagating laminar premixed flame was simulated using ANSYS CHEMKIN-PRO 2020 R2 [41]. The domain extended to 10 cm, and the maximum number of grids allowed was set to 4000. The number of adaptive grid points was set to 50. The adaptive grid control based on the solution gradient and curvature were set to 0.01. The settings correspond to a very dense grid to ensure grid independency. Several mechanisms were used to evaluate the S_L . For ease of discussion, the mechanism by Han et al. [26], Mei et al. (2019) [44], Mei et al. (2021) [14], Nakamura et al. [45], Okafor et al. [23], Otomo et al. [15], Shrestha et al. [22],

and the one presently developed will be referred to as Han Mech, Mei Mech (2019), Mei Mech (2021), Nakamura Mech, Okafor Mech, Otomo Mech, Shrestha Mech, and Present Mech, respectively. Soret effects were not considered in the calculation.

Predictions using Han Mech were close to the present measured S_L and used as the base mechanism. However, Han Mech was found to underestimate and overestimate the measured S_L for lean and rich mixtures, respectively. Reactions with considerably large sensitivity coefficients with respect to the laminar burning velocity at those conditions were identified for various reaction mechanisms [15,23,26,44,45] and are available in Section S.1 of the supplementary materials. Mei Mech (2021) and Shrestha Mech were not included in the sensitivity analysis as those mechanisms were

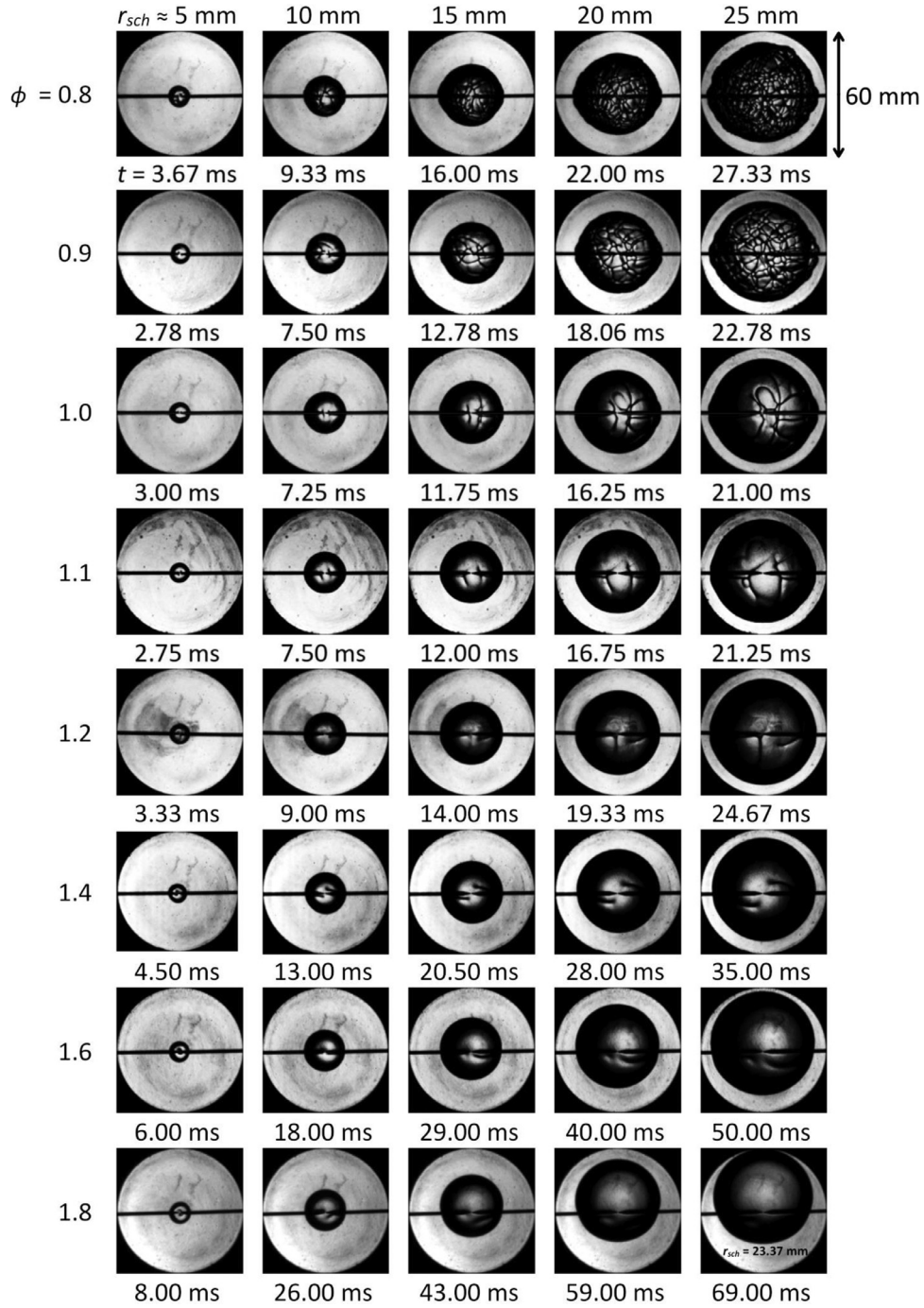


Fig. 2. Schlieren images of $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames for $X_{\text{H}_2} = 0.4$ at 0.5 MPa. The flame radius for $\phi = 1.8$ only reached 23.37 mm before propagating outside the observable range.

only recently published. The rate constants of the identified reactions were considered for revision using data from various studies, and data in the National Institute of Standards and Technology (NIST) database [46]. The rate constants were chosen to optimize the prediction of the measured laminar burning velocity data by the new kinetics mechanism.

The new mechanism (Present Mech) was developed after revising the rate constants of 11 reactions and adding 2 new reactions to the Han Mech, as listed in Table 2. The reaction rates are shown in the modified Arrhenius expression given in Eq. (7); k is the rate constant, A is the pre-exponential factor, T is the temperature, E_a is the activation energy, n is the temperature exponent, and R is

the universal gas constant. The units are in cm, mol, s, cal, and K. The thermodynamic and transport data from Han Mech were used, which originated from Glarborg et al. [49]. Details on the reactions with modified rate constant are available in Section S.2 of the supplementary materials.

$$k = AT^n \exp\left(-\frac{E_a}{RT}\right) \quad (7)$$

As the mechanism was developed for $\text{NH}_3/\text{H}_2/\text{Air}$ flames, carbon-containing species and their corresponding reactions were removed, resulting in a mechanism with 32 species and 165 elementary reactions. Furthermore, Han et al. [26] have identified

Table 2
Summary of the revised rate constants in the Present Mech with their sources.

Reaction	A	n	E _a	Source(s)
1. NH ₃ + OH = NH ₂ + H ₂ O	1.56E+05	2.372	118.9	[47]
2. *H + O ₂ (+M) = HO ₂ (+M)	4.65E+12	0.44	0	[44,48–50]
Low pressure limit	6.37E+20	−1.72	524.8	
TROE	0.5	1E−30	1E+30	
Third body efficiency	H ₂ /2.0/ H ₂ O/14/ O ₂ /0.78/ Ar/0.67/ He/0.8/			
3. H + HO ₂ = H ₂ + O ₂	1.86E+08	1.72	−1157	[51]
4. ^a HNO + O ₂ = NO + HO ₂	2.00E+13	0	15,898	[47,52]
5. HNO + OH = NO + H ₂ O	1.20E+09	1.189	334	[53,54]
6. NH + O ₂ = HNO + O	3.92E+13	0	17,885	[55]
7. NH ₂ + N = N ₂ + H + H	5.00E+13	0	0	[44,56,57]
8. NH ₂ + NH ₂ = N ₂ H ₂ + H ₂	7.83E+11	0	0	[58]
9. NH ₂ + NO = NNH + OH	1.43E+07	1.40	−1777	[59]
10. NH ₂ + NO = N ₂ + H ₂ O	2.70E+17	−1.85	0	[60]
11. ^a NH ₂ + H(+M) = NH ₃ (+M)	1.60E+14	0	0	[53,61]
Low pressure limit	3.6E+22	−1.76	0	
TROE	0.5	1E−30	1E+30	1E+30
	1.20E+12	−0.34	149	[62]
12. NNH + O ₂ = N ₂ + HO ₂	4.38E+13	0	6991	[45,63]
13. OH + H ₂ = H + H ₂ O				

*Revision in the low-pressure limit, TROE, and third body efficiency.

^a New reactions added.

in their mechanism that reactions involving HONO, HNOH, HON, HNO₂, HONO₂, and NO₃ have a negligible influence on the S_L predictions. Those reactions were also found to have a negligible influence in the Present Mech (See Section S.3 of the supplementary material for more details), and therefore the Present Mech can be reduced to 26 species and 119 elementary reactions by removing these species and their corresponding reactions. The detailed and reduced versions of the Present Mech are available as additional materials of this study. It should be noted that all analyses in Section 4 were performed using the detailed version of the Present Mech.

4. Results and discussion

4.1. Experimental and numerical results of the unstretched laminar burning velocity

Figure 4 shows the measured and predicted S_L for NH₃/H₂/Air premixed flames at $X_{H_2} = 0.4$. The uncertainty for the present measurements is based on two standard deviations, while the rest are from their reported uncertainty. The exact values and the standard deviations for the present measurements are available in Section S.4 of the supplementary materials. The present measure-

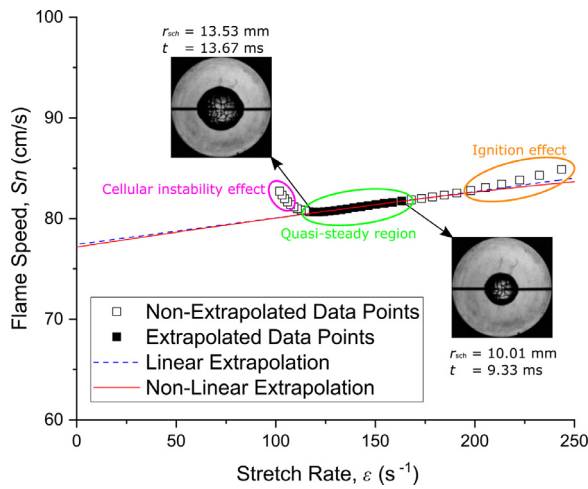


Fig. 3. Flame speed as a function of stretch rate for $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames at $X_{\text{H}_2} = 0.4$, $\phi = 0.8$ under 0.5 MPa.

ments at lean conditions and at 0.5 MPa are not shown in Fig. 4b because they were affected by cellular flame instability and thus considered inaccurate. On the other hand, the figure shows that the non-linear measurements agree with the previously reported measurements [11,27,31,64], while the linear measurements deviate from the non-linear measurements at leaner and richer conditions at 0.1 MPa. The deviation between the linear and non-linear measurements suggests a stronger influence of stretch on the flame speed due to large flame thicknesses and a deviation of the effective Lewis from unity [40] as further discussed in Section 4.4. However, at 0.5 MPa, the linear and non-linear measurements agree more closely due to the smaller flame thickness at the elevated pressure condition.

Figure 4 shows that the prediction by the Present Mech is within the limits of uncertainty for most of the measurements. Other mechanisms also show good estimates, although their predictions start to deviate from limits of measurement uncertainty at certain conditions. For instance, Han Mech, which is the base mechanism for the Present Mech, overestimates S_L at $\phi = 1.5$ and above for 0.1 MPa, and $\phi = 1.1$ and above for 0.5 MPa. Another well-performing mechanism includes the recently developed

Mei Mech (2021), which show good estimates for flames under 0.1 MPa. However, at 0.5 MPa, its predictions are outside the limits of uncertainty in the measurement for $\phi = 1.1$ and above. Comparison between the experimental and numerical results suggests that the Present Mech has improved rich S_L prediction.

To further assess the performance of the Present Mech, reported S_L measurements in the literature at conditions different from those of the current study were compared with predictions by Han Mech and Present Mech. The comparisons are available in Section S.5. of the supplementary material. The comparison suggests that Present Mech satisfactorily predicted the reported S_L measurements of mixture at $X_{\text{H}_2} = 0.4$ under 0.3 MPa (Fig. S9a in the supplementary material) and mixture at $X_{\text{H}_2} = 0.6$ under 0.1 MPa (Fig. S9b in the supplementary material). The Present Mech was also found to perform well in predicting the laminar burning velocity of stoichiometry mixture at various X_{H_2} (Fig. S10 in the supplementary material), and H_2/Air mixtures at various equivalence ratios under 0.1 MPa (Fig. S11 in the supplementary material). However, the Present Mech overestimated the S_L of lean NH_3/Air flames (Fig. S12 in the supplementary material). Discrepancy in the predictions of lean NH_3/Air flames using Present Mech can be attributed to the modification of the rate constants to achieve good S_L prediction for $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames at $X_{\text{H}_2} = 0.4$. More detailed analysis is available in the supplementary material.

4.2. Sensitivity analysis of the unstretched laminar burning velocity

Figure 5 shows the normalized first order sensitivity of S_L to the rate constants of the elementary reactions using the Present Mech. The sensitivity analysis includes reactions with the ten largest sensitivity coefficients from each ϕ . The sensitivity analysis at 0.1 MPa (Fig. 5a) and 0.5 MPa (Fig. 5b) list similar reactions with exception of $\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$ at 0.1 MPa. Some of the reactions listed in the analysis are reactions with modified rate constant in the Present Mech (R1, R2, R8, R9, R11, and R13 in Table 2). It is also observed that N_2H_x reactions ($\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$, $\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_3 + \text{H}$, $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_3$, $\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$) have considerable sensitivity coefficient at rich conditions, supporting the previous findings [14,22]. On the other hand, H_2NO reactions, which are important in low to intermediate temperatures [47,65,66], were not listed in the sensitivity analysis and suggests that H_2NO reactions do not play a significant role in the determination of S_L .

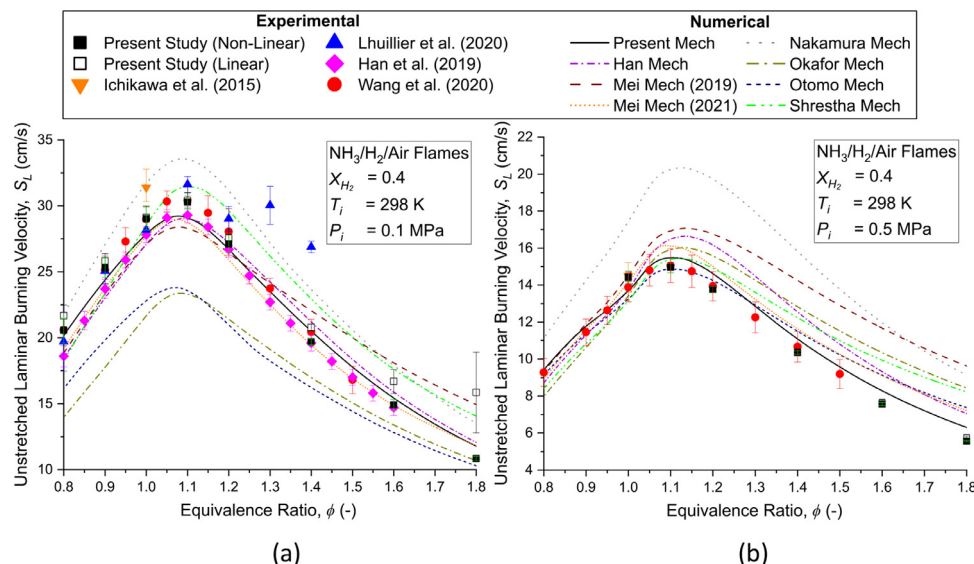


Fig. 4. Unstretched laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames at $X_{\text{H}_2} = 0.4$ for various equivalence ratios under (a) 0.1 MPa and (b) 0.5 MPa.

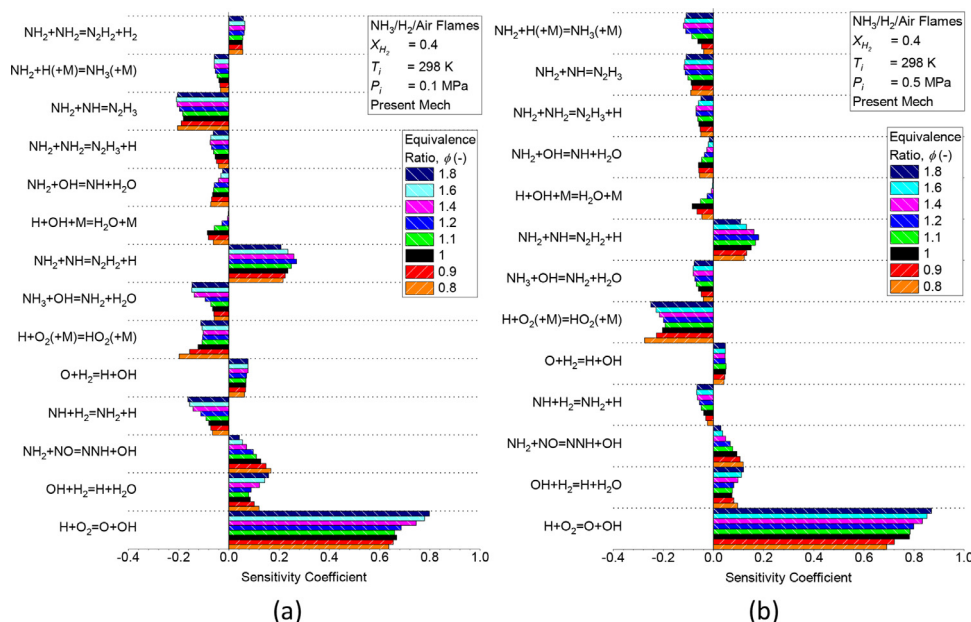


Fig. 5. Sensitivity analysis of the unstretched laminar burning velocity for $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames at $X_{\text{H}_2} = 0.4$ for various equivalence ratios under (a) 0.1 MPa and (b) 0.5 MPa using the detailed Present Mech. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.3. Interactions of NH_3 and H_2 oxidation kinetics

To understand the interactions between the oxidation kinetics of NH_3 and H_2 , the reaction path of the radicals related to H_2 decomposition in NH_3/Air and $\text{NH}_3/\text{H}_2/\text{Air}$ mixtures is shown in Fig. 6. The reaction path is based on the integral of the rate of species production (ROP) over the entire numerical domain (10 cm) using the detailed version of the Present Mech. The blue and black coloured arrows correspond to $\phi = 0.85$ and 1.4, respectively. The arrow thickness shows the magnitude of the molar flux. Species placed at the origin and on the side of the arrow are the reactants, while the species targeted by the arrow is the product. The species beside the arrow have numbers that indicate the percent contribution of the reaction to the total ROP of the targeted species; the blue-coloured number on the left is for $\phi = 0.85$ while the black-coloured number on the right is for $\phi = 1.4$. The diagrams mostly include reactions with percent contribution of at least 5% and the inclusion of reactions below that threshold is only for those considered necessary.

Figure 6 shows that H_2 is decomposed to directly enrich OH and H radicals. The H radicals may oxidize to produce OH, HO_2 , and O radicals or react with NH_x , N_2H_2 , HNO, and HO_2 . The reaction of H atom with NH_3 species, such as the H abstraction reaction $\text{NH}_3 + \text{H} = \text{NH}_2 + \text{H}_2$ is an important NH_3 decomposition step especially at rich conditions. The reactions between H radical and those species produce H_2 that help to recover the H_2 pool. The recovered H_2 may be decomposed to further enrich the H radicals. Since the reactions that help in recovering H_2 also involved NH_3 based species as reactants, it suggests that the NH_3 and H_2 kinetics are coupled through this recovery mechanism. In this study, we named this cycle of H_2 decomposition and recovery/regeneration as H_2 recovery mechanism. The cycle has been previously observed by Otomo et al. [15] and further investigated in this study to better understand the relationship between NH_3 and H_2 .

While it was previously described that the H_2 recovery mechanism helps enrich the H radicals by decomposing the recovered H_2 , the recovery process also requires H consumption. The consumed H atoms might be larger than the amount produced, resulting in an overall lower H radical instead of enrichment. To in-

vestigate this, the normalised integral ROP and rate of consumption (ROC) of H radicals from the kinetics related to H_2 recovery mechanism is shown in Fig. 7. The normalisation used the largest integral ROP/ROC at each condition. NH_3/Air mixtures were used in Fig. 7 because all of the H_2 in NH_3/Air flames originate from the H_2 recovery mechanism and therefore so do the H atoms derived from H_2 decomposition. From Fig. 7, it shows that the lean and stoichiometry flames have small H consumption but high production from the reaction $\text{OH} + \text{H}_2 = \text{H} + \text{OH}$, which results in the net production of H radicals from the H_2 recovery mechanism. On the other hand, more H radicals are consumed by the H_2 recovery mechanism at rich condition, and less H radicals are produced from $\text{OH} + \text{H}_2 = \text{H} + \text{OH}$. The other H formation path, $\text{O} + \text{H}_2 = \text{H} + \text{OH}$, has increased normalised ROP at higher equivalence ratio, which suggests smaller difference in the concentrations between O and OH. It may have to do with the lower percent contribution of the reaction $\text{O} + \text{H}_2 = 2\text{OH}$ that convert O into OH, as seen in Fig. 6. However, despite the increase in ROP from $\text{O} + \text{H}_2 = \text{H} + \text{OH}$, the H consumption at rich condition is much larger than the production and results in the net consumption of the H radicals by the H_2 recovery mechanism.

The reason for the difference in the net production/consumption of the H radicals with the equivalence ratio can be traced to the chain branching reaction, $\text{H} + \text{O}_2 = \text{O} + \text{OH}$. At lean condition, high O_2 concentration in the flame results in significant production of O and OH radicals through the chain branching reaction. Those two radicals are the reactants needed for H_2 decomposition and thus large production of those radicals helps to promote H_2 decomposition. At rich condition, the lack of O_2 discourages the production of O and OH from the chain branching reaction, and subsequently restricts the H_2 decomposition. It results in lower O/H radicals but higher relative concentration of H in the O/H radicals. The increased ammonia content at rich condition also adds more NH_x in the system. The increase of H concentration in O/H radicals and NH_x content in the mixture promote reactions between H and NH_x ; in particular, $\text{NH}_2 + \text{H} = \text{NH} + \text{H}_2$. The reaction $\text{N}_2\text{H}_2 + \text{H} = \text{NNH} + \text{H}_2$ also has significant increase in its H consumption as more NH_2 promotes N_2H_2 production. The promotion of reactions between H

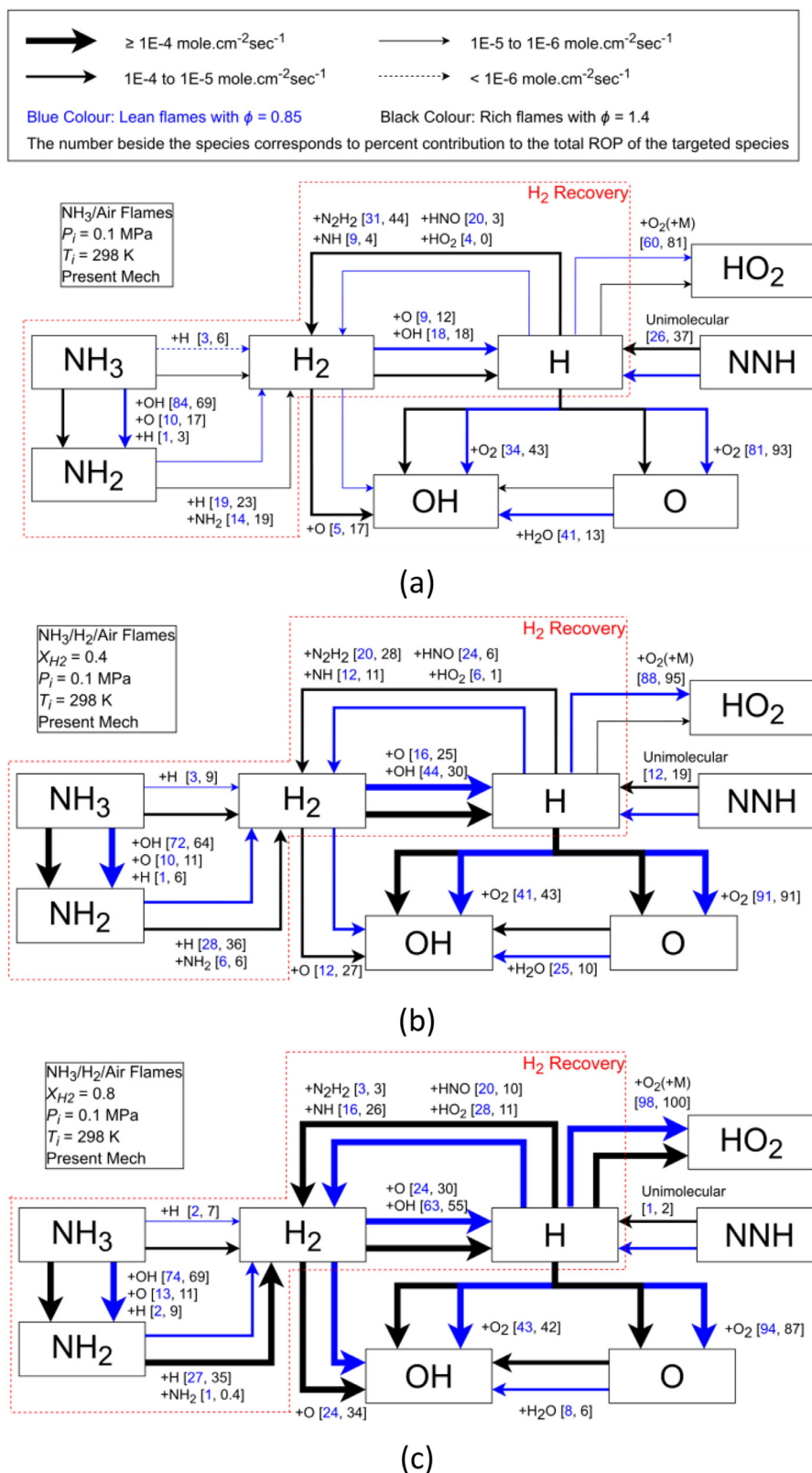


Fig. 6. Integral reaction path for radicals produced through H₂ decomposition for (a) NH₃/Air ($X_{H_2} = 0$), (b) NH₃/H₂/Air at $X_{H_2} = 0.4$, and (c) NH₃/H₂/Air at $X_{H_2} = 0.8$ under 0.1 MPa using the detailed Present Mech (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

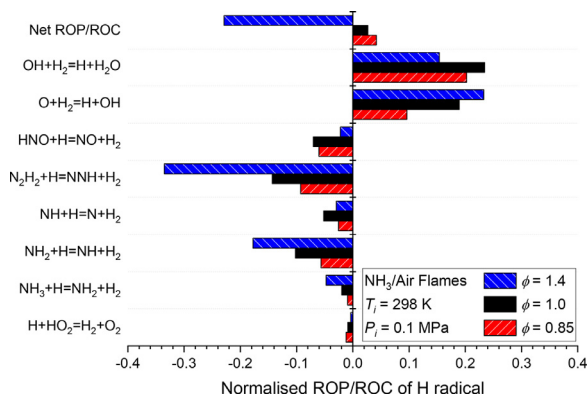


Fig. 7. Normalised integral rate of production/consumption (ROP/ROC) of H radical from reactions related to the H_2 recovery mechanism in NH_3 /Air mixtures under 0.1 MPa for various equivalence ratios using the detailed Present Mech (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

and NH_x or N_2H_2 increases H consumption associated with the H_2 recovery mechanism.

The effect of H_2 content on the H_2 recovery mechanism was also investigated by comparing the reaction path for mixtures at different X_{H_2} , as shown in Fig. 6. The comparison shows that increasing the X_{H_2} results in larger O/H radical production and the rate of H_2 recovered. Looking at the reactions that involve N_2H_x ($NNH = N_2 + H$, $NH_2 + NH_2 = N_2H_2 + H_2$, and $N_2H_2 + H = NNH + H_2$), it shows that higher X_{H_2} reduces their percent contribution towards H and H_2 . The lower contribution means that those reactions have less influence on the O/H radicals. On the other hand, percent contribution from H_2 and NH_x based reactions that do not involve N_2H_x either increases or remains the same with higher X_{H_2} . The difference in the percent contribution between NH_x and N_2H_x reactions suggests different kinetic interactions between various NH_3 based species and H_2 . The reduced importance of N_2H_x species may have to do with the lower NH_3 content at higher X_{H_2} which discourage NH_x production. The available NH_x reacts with H radicals for H_2 recovery, leading to small reactants remaining to produce N_2H_x . At intermediate X_{H_2} , all NH_3 and H_2 based reactions involved in the H_2 recovery mechanism have a considerable contribution to H_2 or H. It suggests that they are equally important and better modelling of those reactions help to improve the flame modelling.

To investigate the effect of pressure on the H_2 recovery mechanism, the ROP/ROC from reactions related to the H_2 recovery mechanism is compared between pressures, as given in Fig. 8. Figure 8 has similar principle as Fig. 7. Observing Fig. 8, it shows that higher pressure results in the decrease of the consumption and production of H radicals at both rich and lean conditions. The decrease can be associated with the promotion of the three-body reaction, $H + O_2(+M) = HO_2(+M)$. The three-body reaction competes for H radicals with the chain-branching reaction $H + O_2 = O + OH$ that reduces the O/H radical pool [7]. Lower O and OH discourages H_2 decomposition and explain the reduced H production with increased pressure. The competition for H radicals also hinders the reactions between H and NH_x that contribute to the recovery of H_2 , explaining the reduced H consumption associated with the H_2 recovery mechanism. The competition caused by the three-body reaction do not significantly alter the net production or consumption of H radical. At lean condition, H_2 recovery mechanism still positively contributes to the overall H radical pool while at rich condition, the H_2 recovery mechanism has negative contribution to the overall H radical pool under high pressure.

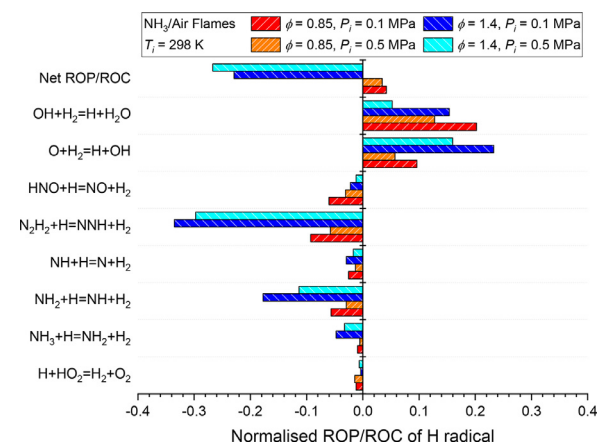


Fig. 8. Normalised integral rate of production/consumption (ROP/ROC) of H radical from reactions related to the H_2 recovery mechanism in NH_3 /Air mixtures under 0.1 and 0.5 MPa for lean and rich mixtures using the detailed Present Mech (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

4.4. The burned gas Markstein length

The L_b measurements and theoretical evaluation for NH_3/H_2 /Air premixed flames at $X_{H_2} = 0.4$ are given in Fig. 9a [11] and 9b, respectively. Lean measurements at 0.5 MPa are not shown in Fig. 9a because, as discussed in Section 4.1, the measurements are affected by cellular instability and considered inaccurate. The results show L_b increased monotonically with ϕ from negative value at $\phi = 0.8$ to positive value at $\phi = 1.8$. Negative L_b at lean condition indicates an increase in the flame speed due to stretching. Similar with the S_L measurements, the discrepancy between linear and non-linear L_b increases with richer and leaner mixtures. Nevertheless, the non-linear results are considered as the more accurate results for similar reason previously stated in Section 4.1. At 0.5 MPa, the absolute L_b value decreases due to the lower flame thickness, which indicates less sensitivity of the flame to stretch. The lower response of the flame to stretch can be seen in the smaller discrepancy between linear and non-linear measurements at elevated pressure.

To understand the reason behind the monotonic trend, a theoretical evaluation based on the asymptotic analysis [67] was performed. The expression used for the analysis is given in Eq. (8), with its first and second terms correspond to the pure stretch and thermo-diffusive effects, respectively. The expression evaluates the theoretical Markstein length, L_{bT} , through its relationship with the effective Lewis number, Le_{eff} , Zeldovich number, Ze , preheat zone thickness, δ_l , and the thermal expansion coefficient, $\sigma = \rho_u/\rho_b$ (ratio of unburned and burned gas densities). Those parameters needed were obtained through calculations using Han Mech in ANSYS CHEMKIN-PRO 2020 R2 [41] and software tools from Colorado State University [68]. The exact value of the parameters used in the theoretical evaluation is available in Section S.6. of the supplementary materials.

$$L_{bT} = \delta_l \left\{ \frac{2}{\sqrt{\sigma} + 1} + \frac{2Ze(Le_{eff} - 1)}{\sigma - 1} \left[\sqrt{\sigma} - 1 - \ln \frac{\sqrt{\sigma} + 1}{2} \right] \right\} \quad (8)$$

Le is defined as the ratio of thermal diffusivity of the mixture and molecular diffusivity of the deficient reactant and indicates the influence of thermodiffusion on L_b [69]. In flames with $Le_{eff} > 1$, where the thermal diffusivity dominates, the flame instability is partially suppressed and resulted in a stable laminar flame. On

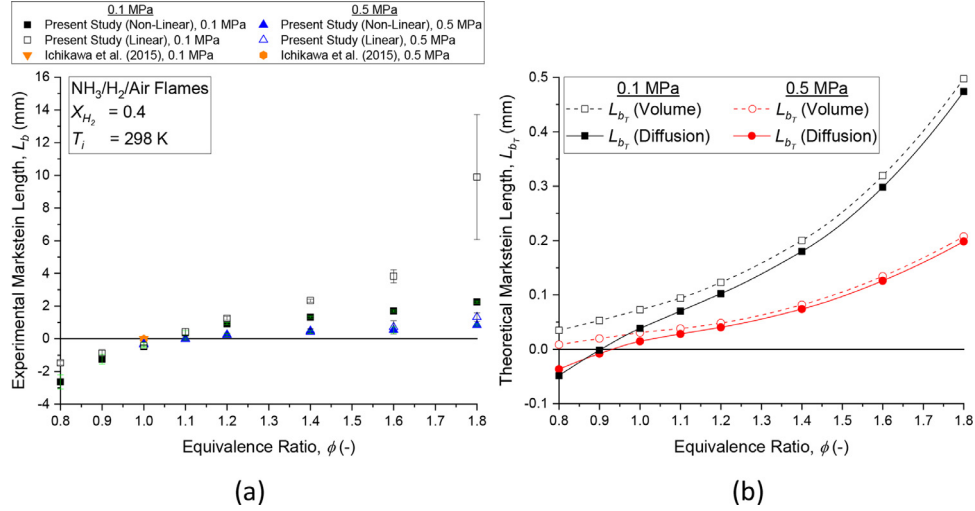


Fig. 9. (a) Experimental and (b) theoretical Markstein length of NH₃/H₂/Air premixed flames at X_{H₂} = 0.4 for various equivalence ratios and pressures.

the other hand, in flames with $Le_{eff} < 1$, the molecular diffusivity dominates and the flames tend to destabilise which may lead to cellular instability [70,71]. To obtain Le_{eff} , Eq. (9) [67] was employed. Le_D and Le_E is the Le for the deficient and excess reactants, respectively. M is the 'mixture strength' and defined in Eq. (10) with Φ as the ratio of the mass excess-to-deficient reactant in the mixture [67]. It was calculated as $\frac{1}{\phi}$ and ϕ for lean and

rich flames, respectively.

$$Le_{eff} = 1 + \frac{(Le_E - 1) + (Le_D - 1)M}{1 + M} \quad (9)$$

$$M = 1 + Ze (\Phi - 1) \quad (10)$$

Le_D and Le_E may correspond to Le of the fuel, Le_{fuel} , or the oxidizer, Le_{O_2} , depending on ϕ . To evaluate the Le of a binary fuel,

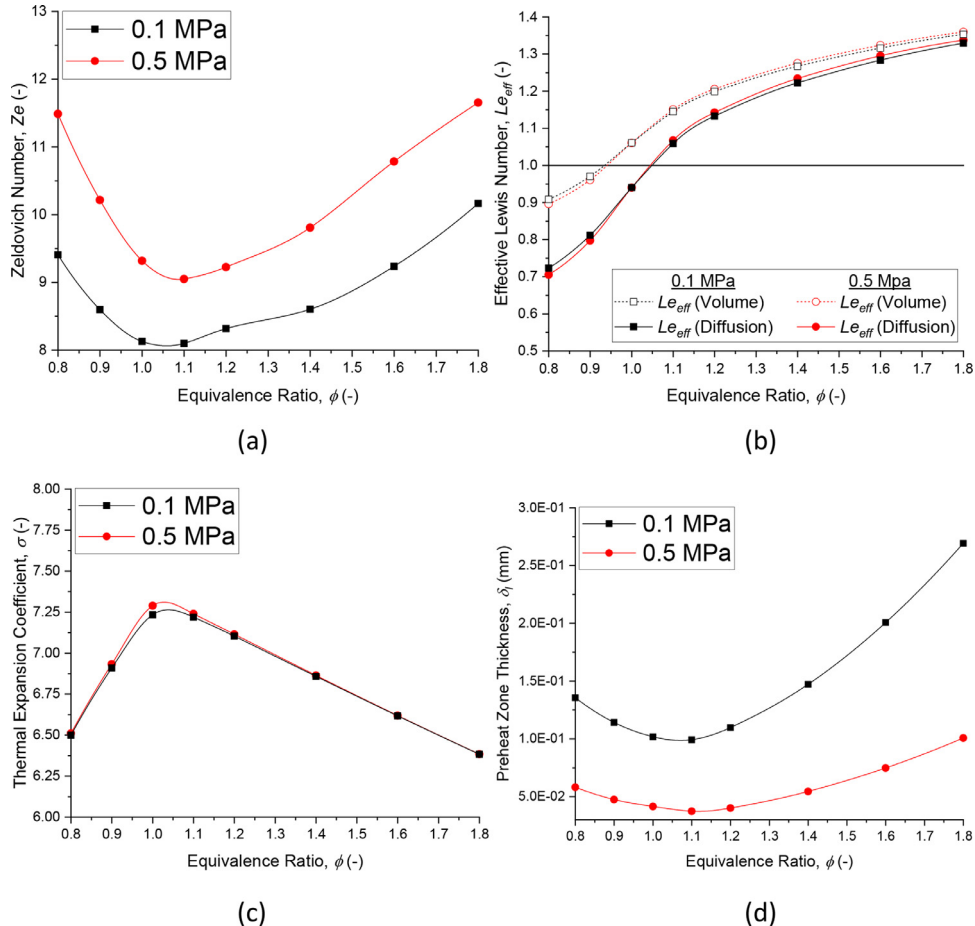


Fig. 10. (a) Zeldovich number, (b) effective Lewis number, (c) thermal expansion coefficient, and (d) preheat zone thickness of NH₃/H₂/Air premixed flames at X_{H₂} = 0.4 for various equivalence ratios and pressures.

relations based on the heat release, volume fraction, and diffusion may be employed [72]. As the heating value of NH_3 is much lower than H_2 [7], only definitions based on volume fraction and diffusion were employed and given in Eqs. (11) and (12) [72,73], respectively. D_T is the mixture's thermal diffusivity while Le_i , D_i , and x_i , are the Lewis number, mass diffusivity, and volume fraction of the component 'i', respectively.

$$Le_{fuel(V)} = x_{\text{NH}_3} Le_{\text{NH}_3} + x_{\text{H}_2} Le_{\text{H}_2} \quad (11)$$

$$Le_{fuel(D)} = \frac{D_T}{x_{\text{NH}_3} D_{\text{NH}_3} + x_{\text{H}_2} D_{\text{H}_2}} \quad (12)$$

Ze is defined as the dimensionless global activation energy of the chemical reaction. Since Ze may express the sensitivity of the burning mass flux to variation of the flame temperature [74,75], it may correlate with the sensitivity of the burning velocity to the modification in flame temperature by flame stretch when $Le \neq 1$ [33]. Ze is calculated using Eq. (13) [76]. T^0 is the inner layer temperature defined as the point on the temperature profile with maximum gradient. T_b and T_u are the adiabatic flame and unburned mixture temperature, respectively.

$$Ze = 4 \frac{T_b - T_u}{T_b - T^0} \quad (13)$$

The stabilizing effect of the flame stretch is associated with the flame thickness. Under the same flame curvature, thicker flames may produce larger stretch rate and thus results in higher stabilizing effect towards the hydrodynamic instability [77]. δ_l was evaluated by employing Eq. (14) [33,42]. λ and c_p is the thermal conductivity and specific heat capacity of the mixture, respectively. S_L used in the theoretical evaluation is based on the Present Mech.

$$\delta_l = \frac{\lambda}{\rho_u c_p S_L} \quad (14)$$

The theoretical Markstien length, L_{bT} , as seen in Fig. 9b, does not quantitatively agree with the experimental results. This is common for theoretical evaluation and has been previously reported by Okafor et al. [33]. Nonetheless, L_{bT} qualitatively agrees with the measurements as it has a monotonic trend with ϕ and may be used to analyse the trend by examining the main parameters used in the theoretical analysis. The evaluated Ze , Le_{eff} , σ , and δ_l are shown in Fig. 10. Ze (Fig. 10a) increased with an increase in pressure and has a non-monotonic trend with ϕ . Le_{eff} (Fig. 10b) has two values since it was evaluated using two different definitions of Le_{fuel} (Lewis number of the binary fuel). Nevertheless, both Le_{eff} (volume) and Le_{eff} (diffusion) have a monotonic trend with ϕ . Le_{eff} slightly changes with pressure since it is a function of Ze and Ze has been shown in Fig. 10a to change with pressure. σ (Fig. 10c) has a slight change with ϕ and pressure, while δ_l (Fig. 10d) considerably changes with ϕ and decreases with an increase in pressure.

By looking at the trend of the parameters and their relationship in Eq. (8), it suggests that the monotonic trend is mainly due to the thermo-diffusive effect that is highly influenced by Le_{eff} and Ze . When $Le_{eff} < 1$, the thermo-diffusive effect has a negative contribution towards L_b and vice versa. In contrast, the pure stretch effect always positively contributes to L_b as σ is always positive in practice [33]. For lean mixture, where $Le_{eff} < 1$, the negative thermo-diffusive effect competes with the positive pure stretch effect and may lead to negative L_b . For rich mixture, where $Le_{eff} > 1$, the positive thermo-diffusive effect adds to the positive pure stretch effect and results in positive L_b . At stoichiometry, the different definition of Le_{fuel} results in either Le_{eff} above or below unity. Since the measured L_b at $\phi = 1.0$ has a negative value, Le_{eff} based on diffusion is the more appropriate definition for the theoretical evaluation.

Ze contributes to L_b by increasing the absolute contribution of the thermo-diffusive effect. The large Ze at lean condition promotes the negative contribution of the thermo-diffusive effect beyond the positive contribution of the pure stretch effect, resulting in the negative L_b . For rich condition, Ze increases at ϕ above 1.1 and promotes the positive contribution of the thermo-diffusive effect, resulting in the large positive L_b .

5. Conclusion

The laminar burning velocity and Markstein length of $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames at intermediate H_2 ratio in the binary fuel have been studied at various pressures and equivalence ratios. A light reaction mechanism optimized using the measured laminar burning velocity has been developed to better model the flames and used to analyse the flame kinetics. From this study, several points may be raised as follows:

- The measured laminar burning velocity is mostly in agreement with the previously reported results and warrants their use to develop a reaction mechanism.
- A reaction mechanism with 32 species and 165 elementary reactions, which can be reduced to 26 species and 119 elementary reactions, has been successfully developed to accurately predict the measured laminar burning velocity. The mechanism was developed by revising the rate constant for reactions with large sensitive coefficient to the laminar burning velocity. The developed mechanism was also found to accurately predict the laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ premixed flames for intermediate H_2 concentration at various equivalence ratios and pressures and has good performance at stoichiometry with various H_2 concentrations and for H_2/Air mixtures at various equivalence ratios at 0.1 MPa.
- A cycle of H_2 decomposition and recovery named H_2 recovery mechanism was observed in the kinetics. The process of recovering H_2 involves NH_3 based species as reactants, suggesting that NH_3 and H_2 kinetics are coupled through this recovery mechanism.
- The recovered H_2 obtained from the H_2 recovery mechanism can be decomposed to enrich H radical, but the process of recovering the H_2 consumes H radical. Through rate of production analysis, it was found that, at lean and stoichiometry conditions, the H_2 recovery mechanism does not consume H radicals as much as it produces H radicals, resulting in its positive contribution to the overall H pool. At rich condition, the H consumption associated with the H_2 recovery mechanism is significantly larger compared to the production, resulting in the negative contribution of H_2 recovery mechanism to the overall H pool. The difference in the contribution of H_2 recovery mechanism to H pool between lean and rich mixtures can be traced to the main chain-branching reaction, $\text{H} + \text{O}_2 = \text{O} + \text{OH}$.
- Higher H_2 concentration in the mixture increases the rate of H_2 recovered and reduces the importance of N_2H_x kinetics. Other reactions that involve NH_x still maintain its importance at high H_2 concentration, suggesting different kinetic interactions between various NH_3 based species and H_2 . At intermediate H_2 concentration, all reactions related to the H_2 recovery mechanism, including N_2H_x reactions, have substantial contribution to H and H_2 . Improving the kinetic modelling of those reactions may lead to more accurate predictions.
- The promotion of $\text{H} + \text{O}_2(+\text{M}) = \text{HO}_2(+\text{M})$ at 0.5 MPa resulted in the competition for H radicals. The competition decreases the consumption and production of H radicals associated with the H_2 recovery mechanism. It does not significantly affect the overall contribution of the H_2 recovery mechanism to H radical.

- The measured burned gas Markstein length has a monotonic trend with the equivalence ratio. Using theoretical evaluation, it was found that the trend is due to the thermo-diffusive effect that mainly changes with the effective Lewis number and Zeldovich number of the flames.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2021.111753](https://doi.org/10.1016/j.combustflame.2021.111753).

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